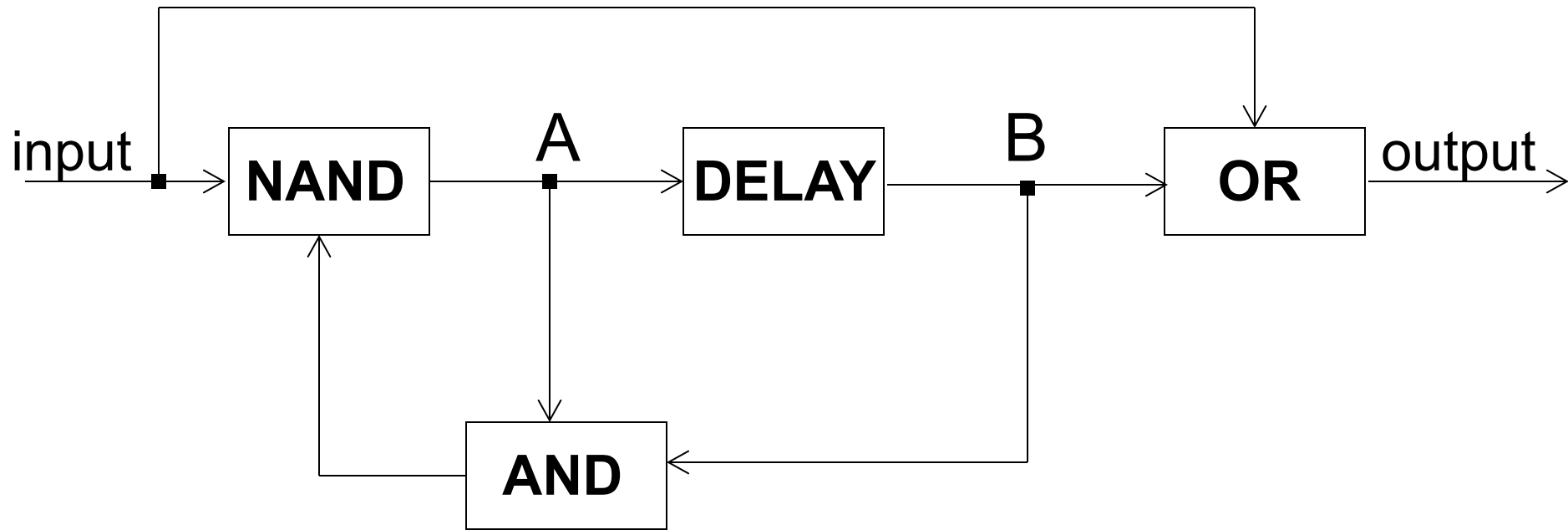


Recap lecture 22

Applications of complementing and incrementing machines, Equivalent machines, Moore equivalent to Mealy, proof, example, Mealy equivalent to Moore, proof, example

Example

Consider the following sequential circuit



The following four types of boxes are used in this circuit

Example continued ...

1. **NAND box (NOT AND)**: For given inputs, it provides the complement of Boolean AND output.
2. **DELAY box (Flip Flop box)**: It delays the transmission of signal along the wire by one step (clock pulse).
3. **OR box**: For given inputs, it provides the Boolean OR output.
4. **AND box**: For the given inputs, it provides the Boolean AND output.

The current in the wire is indicated by 1 and 0 indicates the absence of the current.

Example continued ...

There are two points A and B w.r.t. to which following four states of the machine are identified according to the presence and absence of current at these points *i.e.*

- 1) $q_0(A=0, B=0) \models (0,0)$
- 2) $q_1(A=0, B=1) \models (0,1)$
- 3) $q_2(A=1, B=0) \models (1,0)$
- 4) $q_3(A=1, B=1) \models (1,1)$

Example continued ...

The operation of the circuit is such that the machine changes its state after reading 0 or 1. The transitions are determined using the following relations

$$\text{new B} = \text{old A}$$

$$\text{new A} = (\text{input}) \text{ **NAND** } (\text{old A} \text{ **AND** } \text{old B})$$

$$\text{output} = (\text{input}) \text{ **OR** } (\text{old B})$$

It is to be noted that old A and old B indicate the presence or absence of current at A and B before inputting any letter. Similarly new A and new B indicate the presence or absence of current after reading certain letter.

Example continued ...

At various discrete pulses of a time clock, input is received by the machine and the corresponding output string is generated.

The transition at the state q_0 after reading the letter 0, can be determined, along with the corresponding output character as under

$$\text{new B} = \text{old A} = 0$$

$$\begin{aligned}\text{new A} &= (\text{input}) \mathbf{NAND} (\text{old A} \mathbf{AND} \text{old B}) \\ &= 0 \mathbf{NAND} (0 \mathbf{AND} 0) = 0 \mathbf{NAND} 0 \\ &= 1\end{aligned}$$

$$\text{output} = (\text{input}) \mathbf{OR} (\text{old B}) = 0 \mathbf{OR} 0 = 0$$

Example continued ...

Thus after reading 0 at q_0 new B is 0 and new A is 1 *i.e.* machine will be at state $(1,0) \Rightarrow q_2$ and during this process it's output character will be 0.

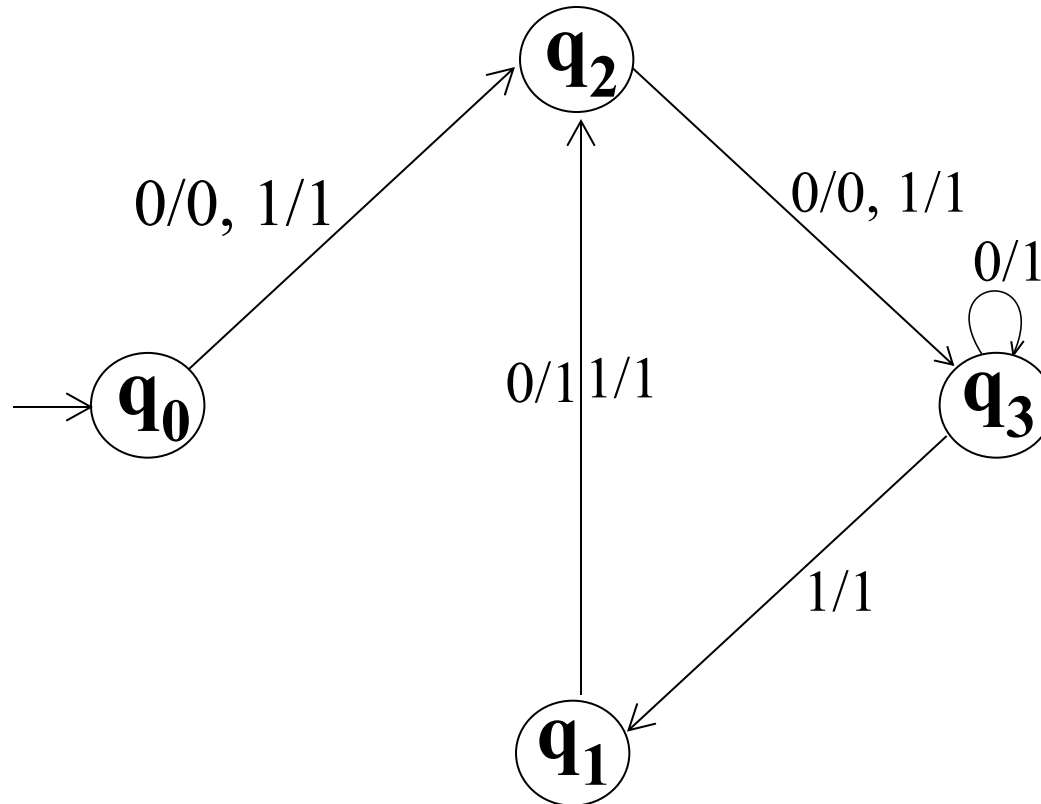
The remaining action of this sequential circuit can be determined as shown by the following suggested transition table of the corresponding Mealy machine

Example continued ...

Old state	Inputting 0		Inputting 1	
	State	Output	State	Output
$q_0 \rightarrow (0,0)$	$(1,0) \rightarrow q_2$	0	$(1,0) \rightarrow q_2$	1
$q_1 \rightarrow (0,1)$	$(1,0) \rightarrow q_2$	1	$(1,0) \rightarrow q_2$	1
$q_2 \rightarrow (1,0)$	$(1,1) \rightarrow q_3$	0	$(1,1) \rightarrow q_3$	1
$q_3 \rightarrow (1,1)$	$(1,1) \rightarrow q_3$	1	$(0,1) \rightarrow q_1$	1

The corresponding transition diagram may be as follows

Example continued ...



Note: It may be noted that if the string 00 is read at any state, it results ending in state q_3 .

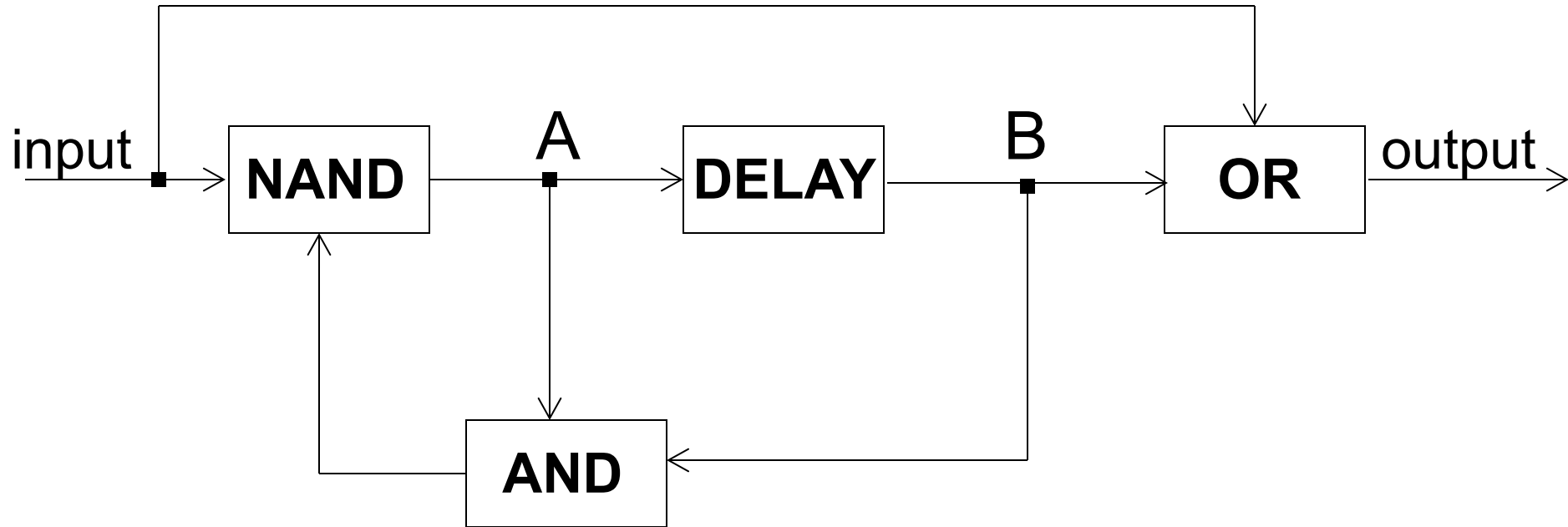
Example continued ...

Running the string 01101110 on the previous machine, the output string can be determined by the following table

Input		0	1	1	0	1	1	1	0
States	q_0	q_2	q_3	q_1	q_2	q_3	q_1	q_2	q_3
output		0	1	1	1	1	1	1	0

Following is a note regarding the sequential circuit under consideration

Note



It is to be noted that in this sequential circuit, delay box plays an important role in introducing four states of the machine.

Summing Up

Mealy machines in terms of sequential circuit